NDUOM SCHOOL OF BUSINESS AND TECHNOLOGY

DESIGN AND IMPLEMENTATION OF 3 ANGELS PHARMACY: AN ONLINE PHARMACY SYSTEM FOR TEMA

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DECLARATION

We hereby declare that this project report is our own original work and has not been submitted to any other institution for the award of a degree or any other qualification. All sources of information used have been duly acknowledged in accordance with the referencing conventions adopted by the Nduom School of Business and Technology (NSBT).

We also confirm that the design and implementation of 3 Angels Pharmacy, the online pharmacy system described in this report, was carried out by us under the guidance of our supervisor, and that any assistance received in the preparation of this report has been acknowledged accordingly.

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Signature: _____ Date: _____

CERTIFICATION / APPROVAL

This is to certify that this project report titled: "Design and Implementation of 3 Angels Pharmacy: An Online Pharmacy System for Tema" was carried out by Vera Akumenuwaa Frimpong (Index No: 10011361) and Juliet Benewaa (Index No: 10011223) of the Department of Information Technology, Nduom School of Business and Technology (NSBT), in partial fulfilment of the requirements for the award of the Bachelor of Science degree in Information and Communication Technology.

The report has been approved for submission to the Department by the undersigned.

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Department of Information Technology, NSBT

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DEDICATION

To our families, for their unwavering love, sacrifice, and encouragement throughout this journey.

To our lecturers and supervisors at NSBT, whose guidance shaped our understanding of technology and research.

And to 3 Angels Pharmacy, Tema, whose story and everyday work inspired the creation of this system.

ACKNOWLEDGEMENTS

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We also wish to acknowledge the entire faculty and staff of the Department of Information Technology at NSBT for the knowledge and skills imparted to us over the course of our studies.

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We are grateful to the staff of 3 Angels Pharmacy, Tema, for their cooperation and for allowing their business to be the basis of this project.

Vera Akumenuwaa Frimpong & Juliet Benewaa

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ABSTRACT

Access to safe, timely, and affordable medication remains a real challenge for many communities in Ghana, where most pharmacies, including 3 Angels Pharmacy in Tema, still depend almost entirely on customers visiting the shop in person. Existing online pharmacy solutions serving Ghana are either large international models built around infrastructure not available locally, or Ghanaian platforms with gaps of their own, including limited price transparency, weak trust mechanisms, and models that risk excluding small, independent pharmacies.

This project presents the design and implementation of an online pharmacy system for 3 Angels Pharmacy, a web-based platform that allows customers to register, browse and search medicines by category, view prices and product details, add items to a cart, check out and pay securely through Paystack, track their orders through a personal profile, and leave reviews of their experience. Pharmacy administrators are supported through an admin dashboard for adding medicines to inventory and viewing customer orders.

The system was built using Vite and React for the frontend, with Supabase used for backend data storage and authentication, and Paystack integrated for payment processing, supporting Mobile Money and card payments widely used in Ghana. The development methodology adopted was Agile, delivered in iterative stages.

The system was evaluated by demonstrating and discussing its working screens against the research questions set out in Chapter One, covering ease of purchase, secure payment, order tracking, admin efficiency, and user acceptance. Results show that the completed system meets its core objectives, successfully combining medicine browsing, secure payment, order visibility, and customer feedback into a single platform for a real, independent pharmacy.

This project contributes a practical, working case study of e-pharmacy system design for an independent pharmacy in Ghana, a context in which academic literature remains limited. Recommendations for future work include full prescription-upload and consultation-booking workflows, production payment activation, expanded delivery logistics, and a native mobile application.

Keywords: Online pharmacy, e-pharmacy, Ghana, Tema, Paystack, Vite, React, Supabase, healthcare access, medicine delivery.

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LIST OF ABBREVIATIONS

AI — Artificial Intelligence

API — Application Programming Interface

APA — American Psychological Association

CSS — Cascading Style Sheets

DFD — Data Flow Diagram

ERD — Entity Relationship Diagram

FK — Foreign Key

GHS — Ghana Cedis (currency)

HTML — HyperText Markup Language

HTTPS — Hypertext Transfer Protocol Secure

ICT — Information and Communication Technology

IT — Information Technology

JS — JavaScript

JSON — JavaScript Object Notation

MoMo — Mobile Money

MTN — Mobile Telephone Network

NEPP — National Electronic Pharmacy Platform

NSBT — Nduom School of Business and Technology

PK — Primary Key

REST — Representational State Transfer

SDK — Software Development Kit

SQL — Structured Query Language

TAM — Technology Acceptance Model

UI — User Interface

URL — Uniform Resource Locator

UX — User Experience

WHO — World Health Organization

WWW — World Wide Web

CHAPTER ONE
INTRODUCTION

1.1 Background of the Study

Many people find it hard to get medicine quickly and easily. They may have to travel a long way to reach a pharmacy, wait in long queues, or visit only during working hours. This can be difficult for sick people, elderly people, or anyone with a busy schedule. At the same time, more people now use phones and the internet to buy things and get services online. This has led to the growth of online pharmacies, which let customers order medicine, talk to a pharmacist, and get their drugs delivered without leaving their homes.

3 Angels Pharmacy is a real pharmacy located in Tema. Like many local pharmacies, it currently depends mainly on customers walking in to buy medicine. This project introduces a web-based online pharmacy system for 3 Angels Pharmacy. The system allows customers to search for medicine, view prices and details, add items to a cart, pay online through Paystack, track their orders, and leave reviews. It also gives pharmacy staff an easy way to manage medicine inventory and view customer orders through an admin dashboard.

The system is designed to work well on phones, tablets, and computers, so that customers can use it from any device they have.

1.2 Statement of the Problem

At the moment, customers of 3 Angels Pharmacy must visit the shop in person for almost everything, including simple tasks like checking if a medicine is in stock. This wastes time and can be stressful for customers who are unwell, busy, or live far from the pharmacy. On the pharmacy side, staff track stock and sales manually, which can lead to mistakes, lost records, and slow service, especially when the shop is busy.

This project addresses these problems by building an online system where customers can shop for medicine, pay, and track orders from anywhere, while staff manage inventory and view customer orders through one organised digital platform.

1.3 Purpose of the Study (Aim and Objectives)

The aim of this project is to design and build a secure, easy-to-use, web-based online pharmacy system for 3 Angels Pharmacy in Tema, so that customers can access pharmaceutical products and services conveniently, and staff can manage the pharmacy's operations more efficiently.

The objectives of the study are to:

1. Build a website where customers can register, log in, and search for medicines.
2. Allow customers to view medicine details and prices, and add items to a shopping cart.
3. Add a secure online payment feature so customers can pay for their orders.
4. Allow customers to track the status of their orders through their profile.
5. Allow customers to leave reviews about their experience with the pharmacy.
6. Build an admin dashboard for pharmacy staff to manage medicine inventory and view customer orders.

1.4 Research Questions

This study will try to answer the following questions:

- How can an online system make it easier for customers to buy medicine from 3 Angels Pharmacy?
- How can the system provide a secure way for customers to pay for medicine online?
- How can customers track the status of their orders after checkout?
- How can the admin dashboard help pharmacy staff manage stock and view customer orders more efficiently?
- How well will customers and staff accept and use this new online system?

1.5 Significance of the Study

This study will benefit 3 Angels Pharmacy by giving it a modern, digital way to reach and serve customers, which can save time, reduce waiting, and reduce mistakes in record keeping. Customers will benefit from a more convenient way to access medicine. The study will also add to knowledge on how small, local pharmacies in Ghana can use technology to improve their

services, which may be useful to other pharmacies, researchers, and students who wish to build similar systems in future.

1.6 Limitations

This project has some limitations. The system depends on customers having a phone or computer with internet access, which not everyone may have. Because of the limited time available for a Level 400 project, the system was not tested with a very large number of users, so results may not fully represent how the system would perform at a much larger scale.

1.7 Delimitations

This project focuses on the core features of the online pharmacy system: registration and login, medicine search and browsing, shopping cart and checkout, online payment through Paystack, order tracking through the customer profile, customer reviews, and an admin dashboard for inventory and order management. The study is limited to 3 Angels Pharmacy in Tema and does not cover other pharmacy branches or national-level pharmacy regulation.

1.8 Organization of the Study

This report has five chapters. Chapter One introduces the study, including the background, problem statement, aim and objectives, research questions, significance, limitations, and delimitations. Chapter Two reviews related literature and existing systems. Chapter Three describes the proposed system, its requirements, design, and the development method used. Chapter Four presents the results of the system and discusses them. Chapter Five gives a summary, conclusion, and recommendations.

CHAPTER TWO
LITERATURE REVIEW

2.1 Introduction

This chapter reviews literature and real-world systems relevant to online pharmacy. It first defines the e-pharmacy concept and its theoretical grounding, then narrows to the state of e-pharmacy research and practice in Ghana and the wider African context, where a genuine and recent research gap exists. It then reviews existing platforms, both international and Ghana-specific, before drawing out six concrete gaps in the current landscape. The chapter closes by showing how 3 Angels Pharmacy's online system is designed to address each of these gaps.

2.2 The E-Pharmacy Concept

E-pharmacy, also called online pharmacy, is an online platform that enables customers to buy medicinal products and services, with convenient delivery to their homes (Ansari et al., 2025). A 2025 systematic review of the online pharmacy literature published in Cureus found that adoption of online pharmacies has grown steadily, driven by lower prices, the availability of medicines not stocked by conventional pharmacies, and the convenience of not having to visit a physical shop (Ansari et al., 2025). The same review organised existing research around three themes: the business models used by online pharmacies, the characteristics of the customers who use them, and customer attitudes and perceptions toward them (Ansari et al., 2025).

A separate systematic review by Pourasad, Norouzi, and Khajouei (2022), published in the Journal of Clinical Research in Paramedical Sciences, looked specifically at what content and functionality an online pharmacy system should have. Drawing on 16 studies, the review found that well-designed online pharmacy systems typically combine product catalogues, secure ordering and payment, and clear delivery and contact information (Pourasad et al., 2022). This gives a research-backed basis for the feature set chosen for 3 Angels Pharmacy's system, namely medicine browsing, cart and checkout, secure payment, order visibility, and a channel for customer feedback.

2.3 E-Pharmacy in Ghana and Africa: A Genuine Research Gap

While e-pharmacy is well studied globally, research specific to Ghana is recent and still limited. Ghana's government piloted a national online pharmacy initiative in December 2021, followed by the launch of the National Electronic Pharmacy Platform (NEPP) in July 2022, described as the first national-scale e-pharmacy platform in Africa (Pharmacy Council of Ghana, n.d.). Eab-Aggrey and Khan (2024), writing in Exploratory Research in Clinical and Social Pharmacy, noted that despite this pioneering rollout, "there is a dearth of such studies in the sub-Saharan African countries" (Eab-Aggrey & Khan, 2024). Their qualitative study of 21 licensed Ghanaian pharmacists found that while pharmacists saw real potential for online pharmacy to improve convenience and privacy, they also raised concerns about poor infrastructure, low health literacy, and inadequate regulation (Eab-Aggrey & Khan, 2024).

A follow-up study by Khan and Eab-Aggrey (2025), published in Digital Health, examined the same Ghanaian e-pharmacy landscape through a health-disparities lens. Pharmacists interviewed for that study described unreliable electricity, poor internet connectivity, and a lack of dependable courier services as real barriers to expanding e-pharmacy in Ghana (Khan & Eab-Aggrey, 2025). Importantly for a project like this one, the same study found that pharmacists feared that, without adequate infrastructure, funding, and regulation, small community pharmacies, often a patient's first point of care, could be displaced by large corporations better able to absorb the cost of digitalisation (Khan & Eab-Aggrey, 2025). This is a directly relevant and citable finding: it establishes that the risk facing independent, community-level pharmacies in Ghana, such as 3 Angels Pharmacy in Tema, is not hypothetical but has been documented by pharmacists themselves.

2.4 Existing Platforms

Existing online pharmacy platforms can be split into two groups for the purpose of this review: international platforms, which are more mature and often assume infrastructure not universally available in Ghana, and Ghana-specific platforms, which are the more direct comparators for this project.

2.4.1 International Platforms

Amazon Pharmacy, and its PillPack service in particular, is a widely documented example of a mature e-pharmacy model. PillPack sorts a customer's medications into date- and time-labelled

packets and ships them monthly, a service aimed at patients managing several regular prescriptions (Amazon, 2025). Amazon Pharmacy relies heavily on integration with U.S. health insurance, at-home delivery infrastructure, and a large logistics network (Amazon, 2025). While instructive as a model of what a mature, well-resourced e-pharmacy platform looks like, this model is not directly transferable to Ghana, where health insurance is not the primary way medicines are paid for and Mobile Money, rather than an insurance card, is the dominant digital payment method.

2.4.2 Ghana-Specific Platforms

Several platforms already operate in the Ghanaian online pharmacy space. DrugNet, a Ghana-based app, allows customers to upload a photo of a prescription and have a pharmacist call back to confirm the order; notably, according to its own app store listing, DrugNet "does not display medications nor prices of medications" upfront, with prices only communicated after a pharmacist calls the customer (DrugNet, n.d.). Aide Chemists operates the Momentum app, which supports Mobile Money and card payment, home delivery or pickup, and, more recently, an AI assistant to help customers understand prescriptions (Aide Chemists, n.d.). mPharma operates a managed-pharmacy network model across several African countries, including Ghana, Nigeria, Kenya, Zambia, and Zimbabwe, serving more than two million people (African Arguments, 2023). At the national level, the Pharmacy Council's NEPP acts as a government-led aggregator connecting licensed pharmacies to patients (Pharmacy Council of Ghana, n.d.). Locally, Top-Up Pharmacy, also based in Tema, already offers retail, wholesale, and delivery services with customer reviews visible on its website (Top-Up Pharmacy, n.d.), making it one of the more directly comparable existing businesses to 3 Angels Pharmacy.

2.5 Gaps Identified in the Literature and Existing Platforms

Drawing directly from the sources reviewed above, six concrete gaps can be identified:

1. Price transparency: DrugNet, one of the more established Ghanaian platforms, does not show medicine prices online; prices are only shared after a phone call (DrugNet, n.d.). This leaves customers unable to browse or compare prices before committing to an order.
2. Infrastructure-appropriate design: Pharmacists in two separate Ghanaian studies identified unreliable power and internet connectivity as real barriers to e-pharmacy

adoption (Eab-Aggrey & Khan, 2024; Khan & Eab-Aggrey, 2025), meaning systems built for Ghana need to be lightweight and resilient rather than assuming constant, high-speed connectivity.

3. Risk of independent pharmacies being excluded: Pharmacists interviewed by Khan and Eab-Aggrey (2025) explicitly feared that community pharmacies could be displaced by larger corporate platforms if digitalisation is not made accessible to them directly, yet most existing Ghanaian solutions (mPharma, NEPP, Aide Chemists) are large chains, franchises, or government aggregators rather than tools an individual neighbourhood pharmacy can use to build its own online presence.
4. Weak trust and feedback mechanisms: none of the Ghana-specific platforms reviewed (DrugNet, Aide Chemists, NEPP) prominently display customer reviews or ratings on their own product pages, even though trust has been shown to be a central factor in e-pharmacy adoption in the wider literature (Ansari et al., 2025).
5. A dearth of Ghana- and Tema-specific documentation: as Eab-Aggrey and Khan (2024) state directly, there is "a dearth of such studies in the sub-Saharan African countries," and the available Ghanaian research focuses on pharmacists' perceptions rather than documenting the design and build of an actual working system for a specific, named pharmacy.
6. Payment methods mismatched to context: international models such as Amazon Pharmacy are built around insurance-linked payment (Amazon, 2025), which does not reflect how most Ghanaians pay for everyday goods and services, where Mobile Money and card payment through local gateways are far more common.

2.6 How 3 Angels Pharmacy Addresses These Gaps

The system built for 3 Angels Pharmacy responds directly to each of the six gaps identified above.

1. Price transparency: unlike DrugNet's call-back model, the Shop page on 3 Angels Pharmacy's website displays the brand, name, description, and price of every medicine upfront, so customers can browse and compare before adding anything to their cart (see Figure 4.6 in Chapter Four).

2. Infrastructure-appropriate design: the system was built as a responsive web application using Vite and React rather than a heavy native app, so it loads on ordinary mobile data and does not require an app store download, reducing the impact of the connectivity constraints identified by Eab-Aggrey and Khan (2024) and Khan and Eab-Aggrey (2025).
3. Support for independent pharmacies: rather than requiring 3 Angels Pharmacy to join a franchise or aggregator network, the system gives the pharmacy its own dedicated storefront and admin dashboard, directly addressing the displacement risk that pharmacists raised in Khan and Eab-Aggrey (2025).
4. Trust and feedback mechanisms: a dedicated Reviews page lets customers leave a star rating and written comment, visible to all visitors, giving 3 Angels Pharmacy a visible trust signal that the reviewed Ghanaian platforms do not prominently offer.
5. Contribution to local documentation: this report itself, describing the design, build, and real screens of a working system for a named Tema pharmacy, is a small but concrete contribution toward closing the documented dearth of Ghana-specific e-pharmacy research (Eab-Aggrey & Khan, 2024).
6. Locally appropriate payment: checkout is handled through Paystack, which supports Mobile Money and card payments widely used in Ghana, rather than an insurance-linked model borrowed from international platforms such as Amazon Pharmacy.

2.7 Summary

This chapter has shown that e-pharmacy is a well-established concept internationally (Ansari et al., 2025; Pourasad et al., 2022), but that research and locally-built systems specific to Ghana remain limited (Eab-Aggrey & Khan, 2024; Khan & Eab-Aggrey, 2025). Existing platforms serving Ghana, whether international models like Amazon Pharmacy or local ones like DrugNet, mPharma, Aide Chemists, and the NEPP, leave real gaps around price transparency, infrastructure-appropriate design, support for independent pharmacies, trust mechanisms, local documentation, and locally appropriate payment. The system built for 3 Angels Pharmacy, detailed in Chapter Three and demonstrated in Chapter Four, was designed specifically to close these gaps for a real, independent pharmacy in Tema.

CHAPTER THREE
PROPOSED SYSTEM / SOFTWARE

3.1 Introduction

This chapter presents the proposed 3 Angels Pharmacy online system, covering its description, functional and non-functional requirements, use case and data design, architecture, interface design, and the development methodology followed.

3.2 System Description

The proposed system is a web-based online pharmacy platform for 3 Angels Pharmacy, located in Tema. The system lets customers create an account, browse medicines by category, view product details and prices, add medicines to a shopping cart, check out and pay online through Paystack, track their order through their profile, and leave a review of their experience.

On the other side, pharmacy administrators use an admin dashboard to add new medicines to the inventory and view customer orders. The website is responsive, meaning it adjusts to fit phones, tablets, and computer screens.

3.3 System Requirements Specification

3.3.1 Functional Requirements

- The system shall allow a customer to register and log in securely.
- The system shall allow a customer to search for and browse medicines by category, including price and description.
- The system shall allow a customer to add medicines to a shopping cart and check out.
- The system shall allow a customer to pay for an order online through Paystack.
- The system shall allow a customer to view their order history and order status from their profile.
- The system shall allow a customer to leave a star rating and written review.
- The system shall allow an admin to add new medicines to the inventory.
- The system shall allow an admin to view customer orders.

3.3.2 Non-Functional Requirements

- The system shall be responsive and usable on phones, tablets, and desktop computers.
- The system shall keep customer data and payments secure.
- The system shall be available and reliable for use at most times.
- The system shall be easy to learn and use for customers and staff with basic computer skills.

3.4 Use Case Diagram

The diagram below shows the main actors of the system — Customer, Pharmacist, and Admin — and the actions each one can perform.

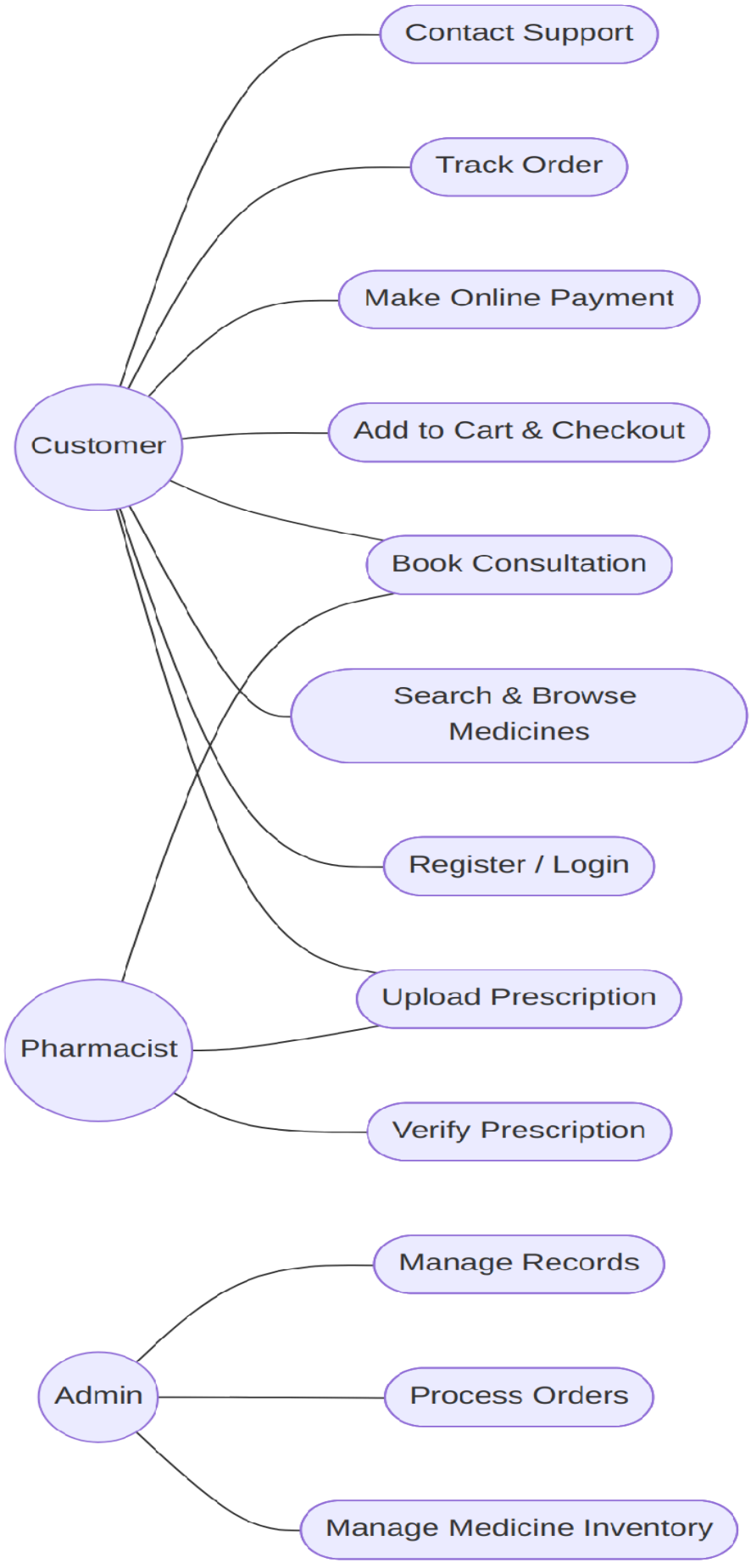

Contact Support
Track Order
Make Online Payment
Customer
Add to Cart & Checkout
Book Consultation
Search & Browse Medicines
Register / Login
Upload Prescription
Pharmacist
Verify Prescription
Manage Records
Admin
Process Orders
Manage Medicine Inventory

Figure 3.1 3 Angels Pharmacy Use Case Diagram

Table 3.1 3 Angels Pharmacy Use Case Descriptions

Use Case	Actor	Description
Search Medicines	Customer	Customer browses or searches the shop by name or category to find a medicine.
Checkout & Pay	Customer	Customer reviews the cart, enters delivery details, and pays securely via Paystack.
Track Order	Customer	Customer views their order history and current status from their profile page.
Leave a Review	Customer	Customer rates their experience and leaves written feedback on the Reviews page.
Manage Inventory	Admin	Admin adds new medicines, with name, brand, category, price, dosage, and stock, to the catalogue.
View Orders	Admin	Admin views the list of customer orders from the admin dashboard.

A typical customer order follows these steps: (1) the customer logs in; (2) the customer browses medicines by category or search and adds items to the cart; (3) the customer proceeds to checkout and enters delivery and contact details; (4) the customer pays through Paystack; (5) the order appears in the customer's profile with a 'pending' status; (6) the admin views the order in the dashboard and processes it; (7) the customer can check their profile at any time to see the order status.

3.5 Architecture Design

The system follows a layered (three-tier) architecture, made up of a presentation layer (the website pages seen by customers and admin), an application layer (the business logic that handles searches, cart, orders, and payments), and a database layer (where all data such as medicines, users, orders, and reviews are stored). The application layer also connects to Paystack for handling payments.

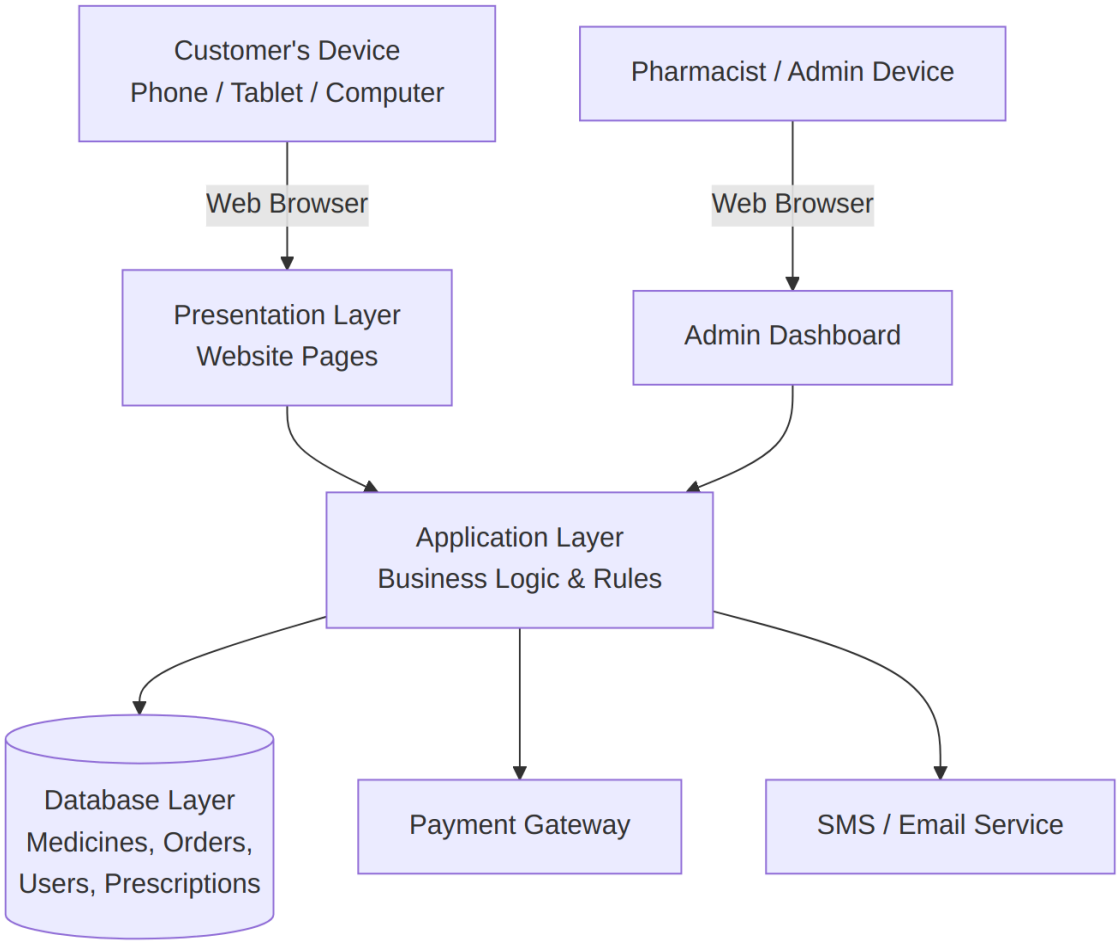


Figure 3.2 High-Level System Architecture

3.6 System Flowchart

The core logic of the system can be summarised in the following simple algorithm for processing an order:

1. Customer logs in or registers.
2. Customer browses medicines and adds items to the cart.
3. Customer proceeds to checkout and enters delivery details.
4. Customer pays through Paystack.
5. System confirms payment and creates the order record with a 'pending' status.
6. Admin views and updates the order from the admin dashboard.
7. Customer checks their profile to see the current order status.

3.7 Data Design

The system's data is organised into the following main tables: Users, Medicines, Orders, Order Items, Payments, and Reviews. Each customer can place many orders and leave many reviews.

Each order can contain many medicines through the Order Items table, and each order is linked to one payment record. The entity-relationship diagram below shows how these tables are connected.

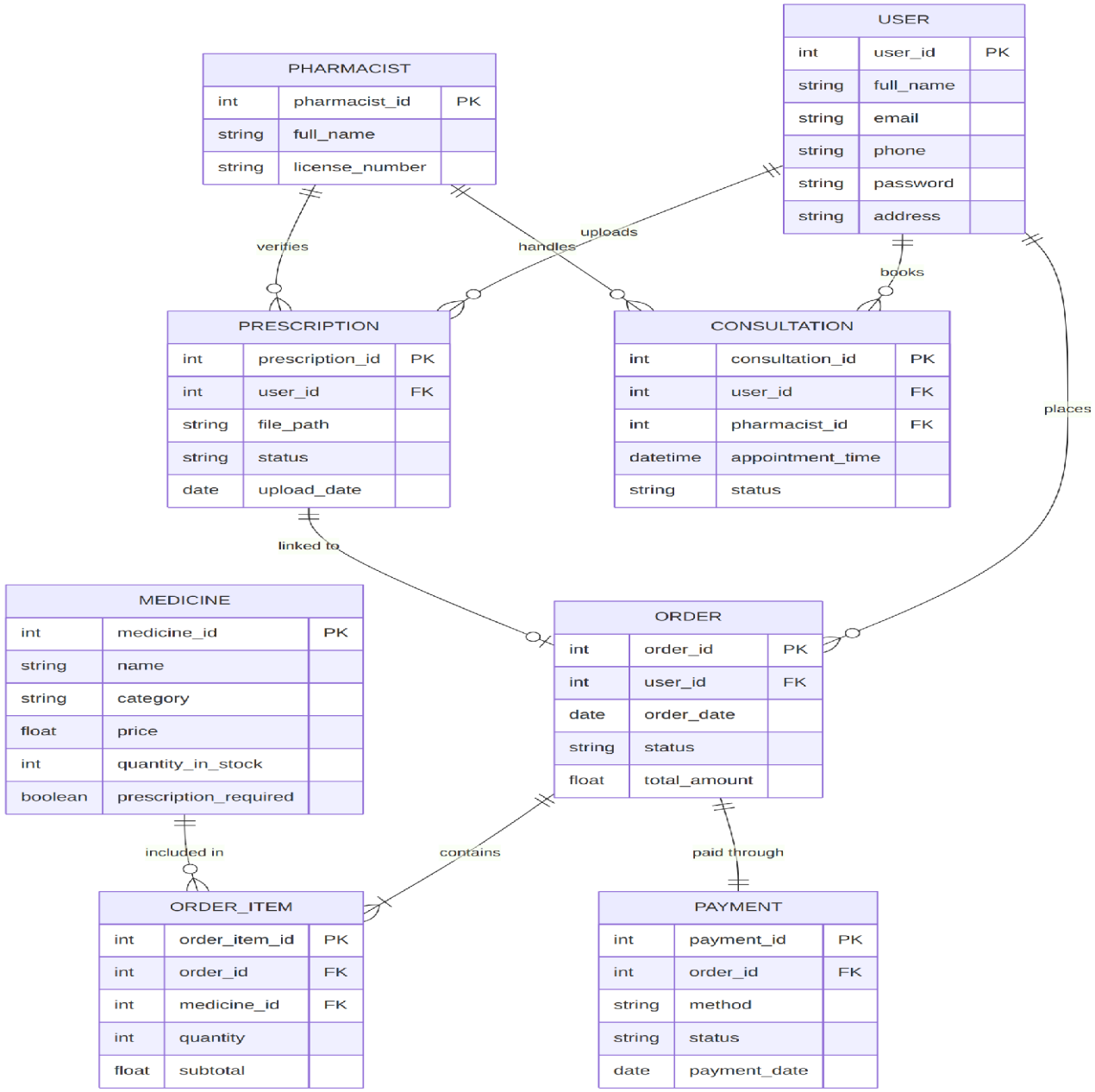


Figure 3.3 3 Angels Pharmacy Entity-Relationship Diagram

The diagram below gives a simple overview of how information moves between the customer, the admin, the payment gateway, and the system.

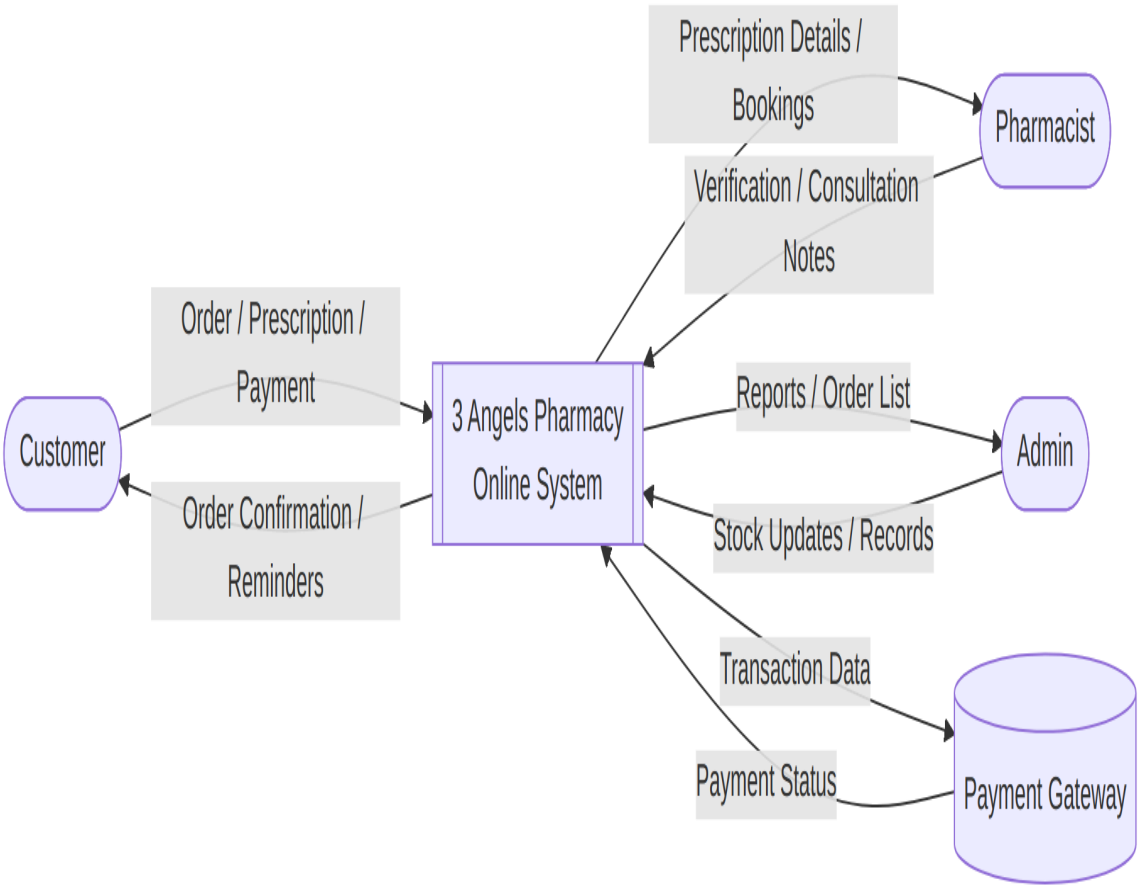


Figure 3.4 Context-Level Data Flow Diagram (DFD Level 0)

3.8 Interface Design

The website includes the following main screens: a Home page with a search bar and feature highlights; a Shop-by-Category section; a Shop page for browsing and filtering medicines; an About & Services page; a Checkout page with delivery details and Paystack payment; a Profile page showing order history; a Reviews page; and an Admin Dashboard for adding medicines and viewing orders. Screens are kept simple and clearly labelled, with a consistent menu, so that customers with basic computer or phone skills can use the system without difficulty.

3.9 Payment System Design

Checkout is handled through Paystack, a payment gateway that supports Mobile Money and card payments widely used in Ghana. At checkout, the customer's order total, made up of the subtotal and delivery fee, is calculated and passed to Paystack for secure processing, after which the order is created with a 'pending' status pending fulfilment by the pharmacy.

3.10 Development Methodology

This project followed the Agile development methodology, building and testing the system in small stages (iterations) rather than all at once. Agile was chosen because it allows the design to be improved gradually based on feedback, which is useful for a project with a fixed academic timeline, since core features (such as registration, search, and cart) could be completed and tested early, while other features (such as the admin dashboard and reviews) were added in later iterations. The system was built using Vite and React, with Supabase used for backend data and authentication, and Paystack for payment processing.

CHAPTER FOUR
RESULTS AND DISCUSSIONS

4.1 Introduction

This chapter presents the results of building the 3 Angels Pharmacy online system. It shows and describes the actual screens produced by the completed system, using screenshots taken from the working website, and discusses how well these results answer the research questions raised in Chapter One.

4.2 Home Page

The home page welcomes customers with the pharmacy's name and location (Tema), a short description of the service, a medicine search bar with an option to search by photo, and quick links to shop medicines or book a consultation. It also highlights same-day delivery across Tema.

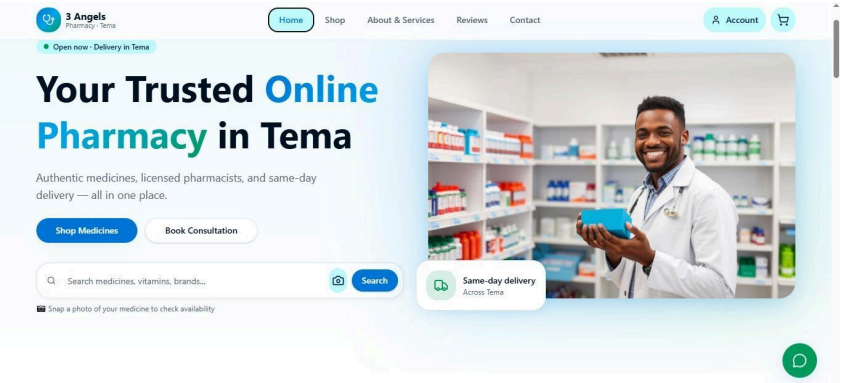


Figure 4.1 Home Page — 3 Angels Pharmacy

4.3 Why Choose 3 Angels (Feature Highlights)

Further down the home page, a features section summarises what the platform offers: buying medicines online, fast delivery, pharmacist consultation, secure payments, prescription upload, and 24/7 support.

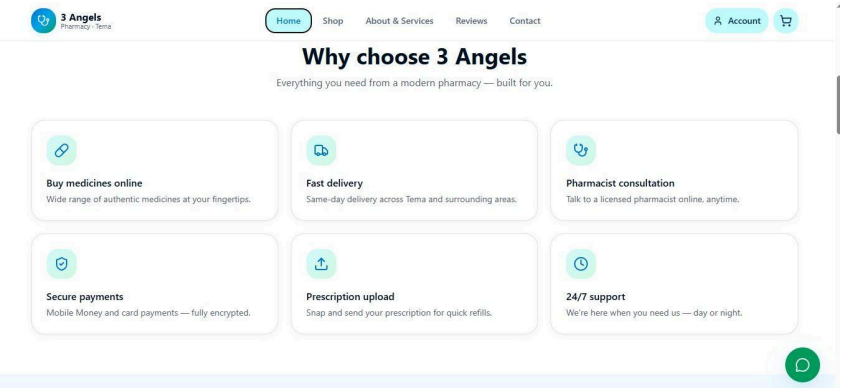


Figure 4.2 "Why Choose 3 Angels" Feature Highlights

4.4 Shop by Category

Customers can shop by category directly from the home page, with categories such as Pain Relief, Vitamins, Baby Care, Skin Care, Diabetes, Supplements, First Aid, and Cold & Flu.

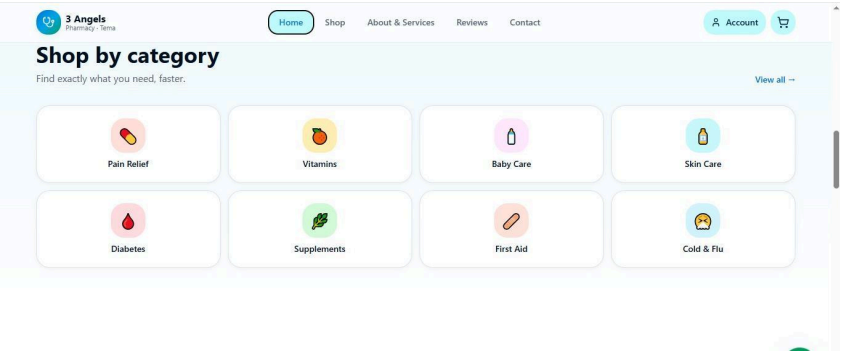


Figure 4.3 Shop by Category Section

4.5 About & Services Page

The About & Services page tells customers about the pharmacy's story, mission, and vision, and lists the services offered, including online pharmacist consultation, prescription refill, medication delivery, health advice, prescription upload, and appointment booking.

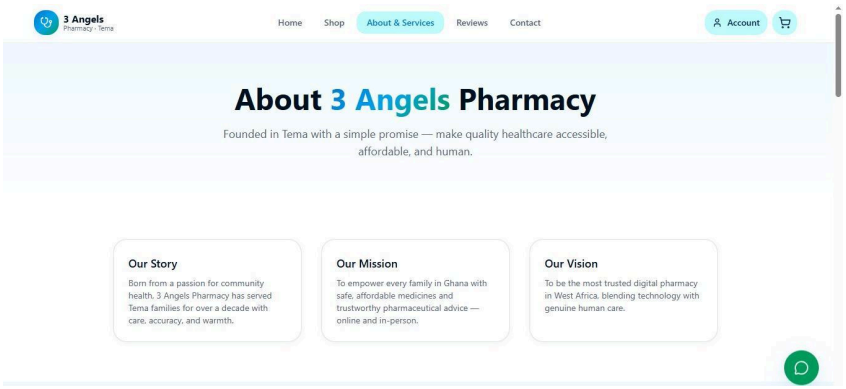


Figure 4.4 About 3 Angels Pharmacy

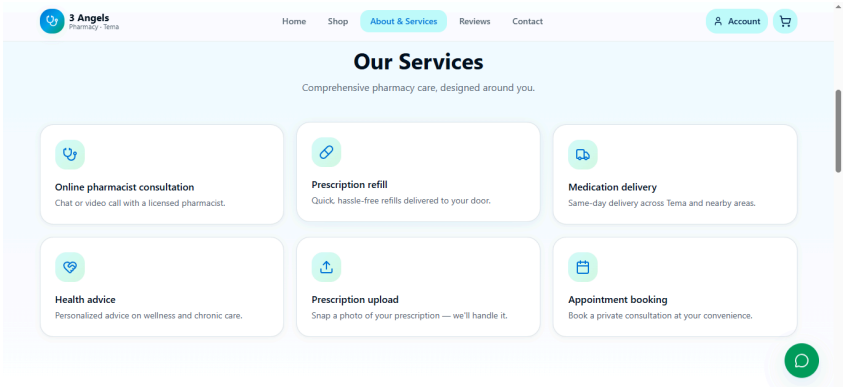


Figure 4.5 Our Services Section

4.6 Shop / Medicine Catalogue

The Shop page lists all available medicines with filters for each category, a search bar, and a sort option. Each product card shows the brand, name, a short description, and the price in Ghana cedis (GH₵), with an 'Add' button for quickly adding the item to the cart.

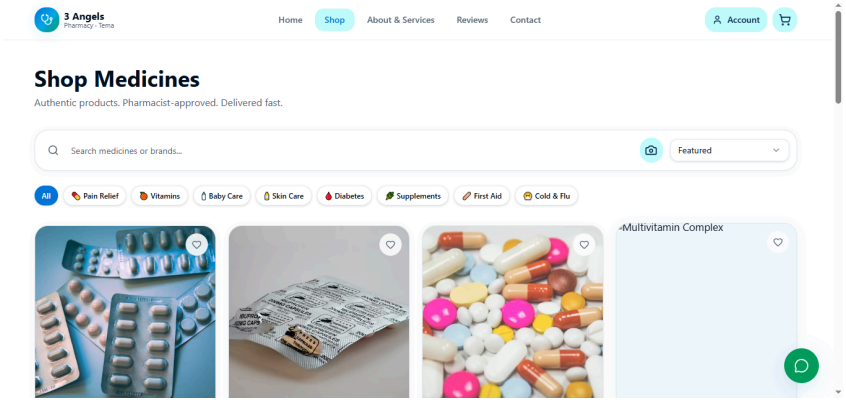


Figure 4.6 Shop Medicines Page with Category Filters

4.7 Checkout and Online Payment

At checkout, the customer enters their delivery and contact details, and sees an order summary showing the subtotal, delivery fee, and total. Payment is completed securely through Paystack, which supports Mobile Money and card payments.

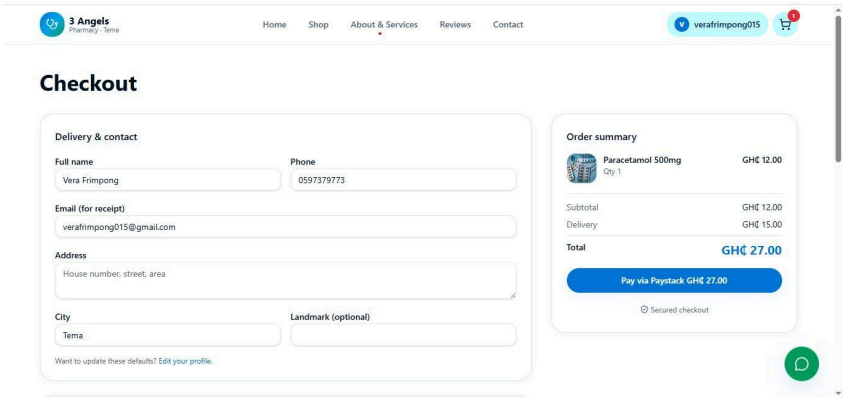


Figure 4.7 Checkout Page with Paystack Payment

4.8 Customer Profile and Order Tracking

The Profile page lets a customer manage their personal details and shows a list of 'My orders', including the order reference number, date, number of items, cost, and current status.

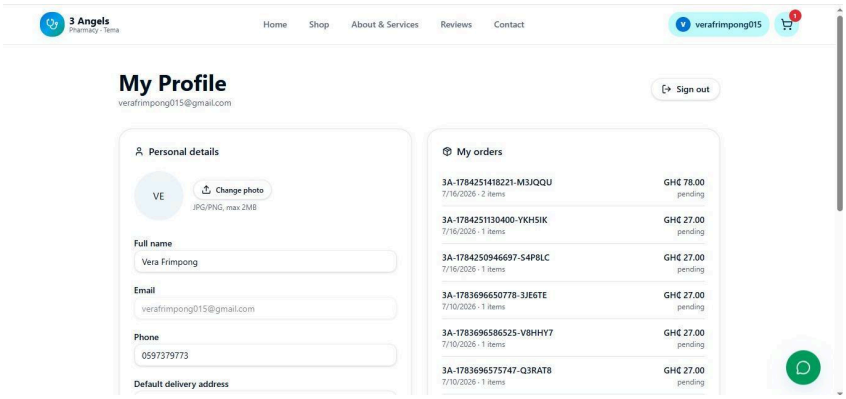


Figure 4.8 Customer Profile Page with Order History

4.9 Customer Reviews

The Reviews page displays customer ratings and comments about the pharmacy, along with an overall average rating. Customers can leave their own review by entering their name, a star rating, and a written comment.

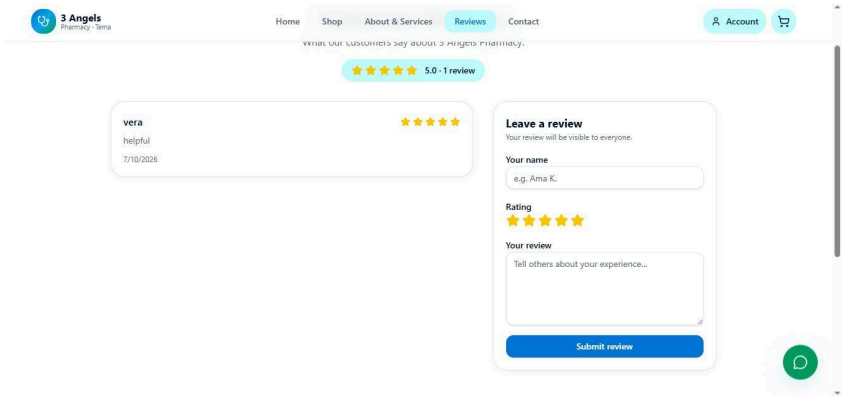


Figure 4.9 Customer Reviews Page

4.10 Admin Dashboard

The admin dashboard gives pharmacy staff a form to add a new drug to the inventory, capturing the name, brand, category, dosage, price, stock level, description, and an optional image. The dashboard also has a 'Customer Orders' panel where admin staff can view and refresh the list of orders placed by customers.

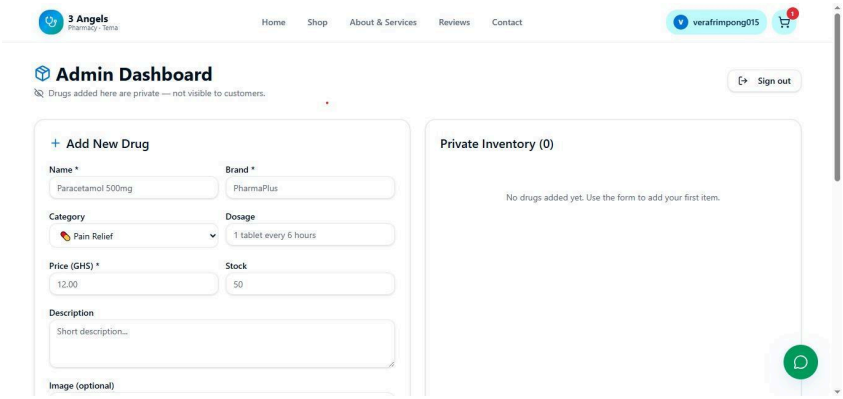


Figure 4.10 Admin Dashboard — Add New Drug Form

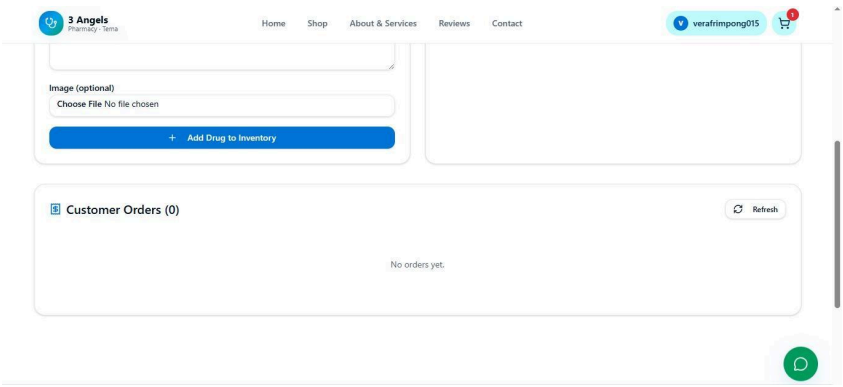


Figure 4.11 Admin Dashboard — Customer Orders Panel

4.11 Discussion of Results

The results are discussed below in relation to the research questions set out in Chapter One.

On how an online system can make it easier for customers to buy medicine, the home page, shop-by-category section, and shop page show that customers can find and order medicine without visiting the pharmacy physically, directly addressing the first research question.

On providing a secure way to pay online, the checkout page demonstrates that customers can complete a purchase using Paystack, which supports Mobile Money and card payments, addressing the second research question with a payment method well suited to the Ghanaian market.

On tracking the status of orders, the Profile page's 'My orders' section shows that customers can view their order history and status at any time, answering the third research question.

On helping pharmacy staff manage stock and view orders, the admin dashboard's 'Add New Drug' form and 'Customer Orders' panel show that inventory and orders can be handled from one screen instead of paper records, which answers the fourth research question and supports more efficient pharmacy operations.

On the level of acceptance of the system by customers and staff, the Reviews page already shows early positive feedback, with an initial 5-star rating recorded from a test user. A larger, more structured round of user testing and feedback would give a fuller picture of acceptance and is recommended as future work.

It is worth noting that some features described at the design stage in Chapter Three, such as a dedicated prescription-upload workflow and a full consultation-booking calendar, are currently presented on the About & Services page as offered services, while the interactive booking and verification screens for these features were still being finalised at the time of writing. This should be completed, tested, and documented with screenshots before final submission if it forms part of the finished system.

4.12 Summary

This chapter has presented the completed screens of the 3 Angels Pharmacy online system and discussed how they answer the research questions set out in Chapter One. The system demonstrates a working, integrated pharmacy platform covering browsing, secure payment, order tracking, and customer feedback.

CHAPTER FIVE
CONCLUSION AND RECOMMENDATIONS

5.1 Summary of Key Findings

This project set out to design and build a web-based online pharmacy system for 3 Angels Pharmacy in Tema, to make it easier for customers to access medicine, and to help pharmacy staff manage stock and orders more efficiently. The completed system allows customers to register, browse medicines by category, add items to a cart, pay online via Paystack, track their orders through their profile, and leave reviews, while giving pharmacy staff an admin dashboard to add medicines to inventory and view customer orders. The results in Chapter Four show that the system meets the core objectives and functional requirements set out in Chapters One and Three, and that it addresses weaknesses identified in existing systems reviewed in Chapter Two, particularly around combining ordering, payment, order tracking, and customer feedback in one platform.

5.2 Conclusion

The study concludes that an integrated online pharmacy platform is both achievable and beneficial for a local pharmacy such as 3 Angels Pharmacy. By bringing together medicine browsing, secure Paystack payment, order tracking, customer reviews, and an administrative dashboard into a single system, the project demonstrates a practical and more convenient alternative to relying solely on in-person visits and manual record keeping. While the system was built and tested within the constraints of a Level 400 academic project, the core modules work together to meet the aim and objectives stated at the outset of the study.

5.3 Recommendations

Based on the findings of this project, the following recommendations are made for 3 Angels Pharmacy and for future developers who may extend this work:

1. The system should be tested with a larger group of real customers and pharmacy staff over a longer period to better evaluate usability and acceptance.
2. The prescription upload and consultation booking features listed on the Services page should be fully built out with working screens, tested, and documented.
3. A dedicated delivery/logistics feature, or partnership with a courier service, should be added to track physical delivery of medicines in real time.
4. The admin dashboard should be extended to allow staff to update order status directly, rather than only viewing orders.
5. The pharmacy should ensure the system meets any relevant national regulatory requirements for the online sale and delivery of medicines before public launch.
6. Future versions could include a native mobile app, in addition to the responsive website, to further improve convenience for customers.
7. Regular backups and security reviews should be carried out to protect customer data and payment information.

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